

INVESTIGATION REPORT

Issue 1016: Write-back contention flag in the EX stage

openhwgroup/cv32e40p

Document	RTL-TRIAGE-CV32E40P-1016
Disposition	Dead code, proven
Record date	2026-09-17
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40p#1016
Issue filed	2024-07-02
Issue type	Coverage hole referred for disposition
Affects	cv32e40p
Configuration	RTL 6033d2b, COREV_PULP=1 FPU=1 (the upstream PULP_F0 configuration), top cv32e40p_formal_top, yosys-slang frontend
Disposition	Dead code, proven
Severity	Not assessed
Agreement	Same disposition as the independent human record.
Evidence	4 cited, 4 present

Disposition meaning. The code cannot be reached in any state permitted by the stated model and assumptions, established by an unbounded proof.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

The uncovered expression row of the if at cv32e40p_ex_stage.sv:211 is unreachable. Line 205 opens the branch taken when an APU result is valid and is classified single cycle or multi cycle; line 211 inside that branch raises wb_contention when regfile_alu_we_i is set and apu_en_i is not. The row never exercised by regression is the one where apu_en_i and regfile_alu_we_i are both 1 inside that branch, because the decoder never enables the APU and an ALU register-file write for the same instruction in the same cycle. An unbounded proof (abc pdr, top cv32e40p_formal_top, COREV_PULP=1 FPU=1) of that exact row passes. A bounded cover run reaches the enclosing branch at step 12, so the assertion is not vacuous. This is the same property as the upstream assert_unreachable_ex_211, checked with a different tool chain.

3 Conditions

1. legal OBI handshakes on both the instruction and the data interface
2. scan_cg_en_i=0
3. no CSR write to the hardware-loop CSRs (0xCC0, 0xCC1, 0xCC2, 0xCC4, 0xCC5, 0xCC6)
4. clock gates modelled as transparent
5. every trace starts in reset
6. instruction memory contents otherwise unconstrained

Reproduced from the assumes that are enabled in the upstream formal environment at cores/cv32e40p/scripts/formal/src, plus the two modelling choices the open-source flow requires.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Waiver text, to be applied to the coverage exclusion

cv32e40p_ex_stage.sv:211 (wb_contention inside the APU write-back branch opened at line 205): the expression row with apu_en_i=1 and regfile_alu_we_i=1 is formally proven unreachable (abc pdr, unbounded, top cv32e40p_formal_top, COREV_PULP=1 FPU=1, RTL 6033d2b). Assumptions: legal OBI handshakes on both interfaces, scan_cg_en_i=0, no CSR write to the hardware-loop CSRs, transparent clock gates. Evidence: runs/v1-cv32e40p-1016/formal/p_assert/sby.log (PASS) and formal/c_antecedent/sby.log (enclosing branch reached at step 12). Exclude that row only; the remaining rows of line 211 are reachable.

Checker used

checker_1016.sv

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	proof	formal/p_assert/sby.log	present	abc pdr PASS (COREV_PULP=1 FPU=1): apu_en_i and regfile_alu_we_i are never both set inside the APU write-back branch of line 205
E2	proof	formal/p_assert/result.json	present	machine-readable status record for the proof job
E3	proof	formal/c_antecedent/sby.log	present	smtbmc/yices cover PASS (depth 40): the line 205 branch is reached at step 12, so a_ex211 is not vacuous
E4	proof	formal/c_antecedent/result.json	present	machine-readable status record for the cover job

Paths are relative to runs/v1-cv32e40p-1016. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.

7 Limitations and scope

- The proof establishes unreachable under the model and the environment assumptions named in Status and Conditions. It does not establish that the surrounding logic is correct.
- This is an internal investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Public demonstration runs on open cores are not client engagements and not production verification results.

8 Provenance and measurement

Run directory	runs/v1-cv32e40p-1016
Inputs shown	title, body(links stripped), labels
Human record	Dolphin Design 2024: waived in v2; reported proven unreachable with Siemens Questa Static formal, assertion assert_unreachable_ex_211
Assessed confidence	0.92. This is the investigator's own estimate. It is not a measurement and not a reliability figure.

Iterations	1
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